ClusterFusion

Project

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"Nowadays, there’s too much information to read. Text summarization helps by quickly picking out the important points from long documents. It saves time and makes things easier to understand."Text summarization refers to the technique of shortening long pieces of text. The intention is to create a coherent and fluent summary having only the main points outlined in the document.

Relationship to Machine Learning :
Machine Learning (ML) is a branch of artificial intelligence (AI) that enables computers to learn from data and make predictions or decisions without being explicitly programmed , Machine learning comes into play when we use advanced summarization techniques like:
Unsupervised Learning: Identifying patterns in the text without labeled data (e.g., clustering important sentences.

NLP in Text Summarization:
Text summarization is a core application of NLP, where algorithms are designed to process and analyze human language (text) to generate summaries. Techniques like tokenization, stopword removal, and sentence scoring

"Good morning everyone. My name is morshaline, and I am presenting my project under the guidance of my supervisor, Dr.sumon ahmed. The topic of my project is Text Summarization."

Overview

Text Summarization is the process of condensing a large text into a shorter version while retaining its key information. It helps extract the most important points from lengthy documents, making them easier to read and understand . The process involves analyzing the text, identifying important sentences or keywords, and generating a concise summary.

Steps of My Project/Key Features

Text Preprocessing:

- Breaking the document into sentences or words (**Tokenization**).
- Removing unnecessary words (**Stopwords**) and normalizing the text.

Text Summarization and the Gaussian Mixture Model (GMM) are not directly related, but GMM can be used in text processing, especially in text summarization tasks. GMM is primarily a clustering method that divides data into different groups GMM uses soft clustering, meaning that each sentence can be related to multiple clusters. This increases the chances of important sentences being assigned to relevant clusters and selected for summarization.

we have use of 3 type of algorhtims
Word Frequency Algorithm: This approach ranks sentences based on the frequency of important words in each sentence. The sentences with higher frequencies of essential words are considered more relevant and are included in the summary

word frequncy algorthims 5 steps implementation

1. Create the word frequency table

2. Tokenization: Breaking the document into sentences/words.

3. Score the sentences: Term frequency : score a sentence by its words, adding the frequency of every non-stop word in a sentence.

4.Find the threshold : we are considering the average score of the sentences as a threshold.

5.Generate the summary :Select a sentence for a summarization, If the sentence score is more than the average score.

- **Tokenization**: The text is divided into sentences.
- **Stop Word Removal**: Common, less meaningful words like "the," "and," or "is" are removed.
- **Word Frequency Analysis**: The tool calculates the importance of words by counting their frequency.
- **Sentence Scoring**: Each sentence is scored based on its meaningful words' frequency.
- **Summary Generation**: Sentences with scores above a calculated threshold are selected for the summary.

Key Features:

- Uses **TF-IDF** logic to identify important words.
- Avoids unimportant or repetitive sentences.
- Implements algorithms for punctuation removal, word conversion to lowercase, and sentence scoring.

```
double tf(const vector<string>& doc, const string& term) {
        double count = 0;
        for (const auto& word : doc) {
                if (term == word) {
                        count++;
                }
        }
        return count / doc.size();
```

```
double idf(const vector<vector<string>>& docs, const string& term) {
        double n = 0;
        for (const auto& doc : docs) {
                if (find(doc.begin(), doc.end(), term) != doc.end()) {
                        n++;
                }
        }
        if (n == 0) {
                return 0;
        }
        return log(docs.size() / n);
}
```

Key Features:

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```
void scoreSentences() {
        for (const string& sentence :
sentences) {
            double sentenceScore = 0;
            int noOfTotalWords =
spaceCounter(sentence) + 1;
            int noOfNonStopWords =
countNonStopWords(sentence);

            sentenceScore =
static_cast<double>(noOfNonStopWords) /
noOfTotalWords;
            sentencesScore[sentence] =
sentenceScore;
        }
    }
```

Key Features:

Extracts important information using TF-IDF.

Avoids unimportant or repetitive content.

Processes punctuation, case normalization, and sentence scoring.

Summary:

Freelance writer Sng Ler Jun has been taking on more writing assignments since March to earn extra money for his family and himself.

Battling pandemic fatigue: Some feel burned out as the Covid-19 outbreak drags on with no end in sight.

Application

1. **News Aggregation**
2. **Search Engines**
3. **Educational Tools**
4. **Customer Support**
5. **Social Media**
6. **Healthcare**
7. **E-Learning and AI Assistant**

Next Key Features :

- Text Rank Algorithms
- Direct Method Algorithm

http://github.com/morshaline1504